Robust Deepfake Detection Under Real-World Perturbations

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***Abstract*—This paper studies robust detection of manipulated face videos with a lightweight network. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise.**

***Index Terms*—deepfake detection, forensics, robustness**

I. Introduction

Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise.

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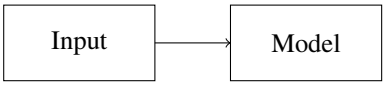


Fig. 1. Overview of the proposed pipeline.

II. Method

Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. The loss is

$$L = \sum_{i} (y_i - \hat{y}_i)^2 \tag{1}$$

and the score is

$$s = \sigma(w^T x + b). \tag{2}$$

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TABLE I
COMPARISON WITH PRIOR METHODS

Method	Acc	AUC
hline Baseline	81.2	0.85 Ours
93.7	0.96	
hline		

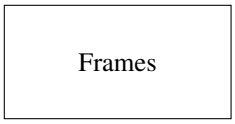


Fig. 2. Example frames.

III. EXPERIMENTS

Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise.

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IV. CONCLUSION

Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise.

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